EXP NO: 1 STUDY OF VARIOUS NETWORK COMMANDS USED IN LINUX AND WINDOWS

AIM:
To study the various networking commands used in linux and windows

1. arp -a

   syntax: arp -a

2. hostname

   Syntax: hostname

3. ipconfig

   syntax: ipconfig/all

4. nbtstat

   syntax: nbstat -a <hostname>

5. netstat

   Syntax: netstat

6. nslookup

   Syntax: nslookup www.google.com

7. pathping

   Syntax: pathping <hostname | IP>

8. ping

   syntax: ping localhost

9. route

   Syntax: route print (windows)

   ip route (linux)

   o/p displays the IP routing table.

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C:\Users\Ramya>arp -a

Interface: 192.168.56.1 --- 0xc
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.0.135 --- 0x10
  Internet Address      Physical Address      Type
  192.168.0.1           e0-1c-fc-e5-47-c6     dynamic
  192.168.0.27          ca-7b-57-b2-7d-22     dynamic
  192.168.0.71          80-47-86-63-57-c8     dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
C:\Users\Ramya>hostname
DESKTOP-0NRMEE7
```

```
C:\Users\Ramya>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:7768         DESKTOP-0NRMEE7:26733  ESTABLISHED
  TCP    127.0.0.1:26733        DESKTOP-0NRMEE7:7768   ESTABLISHED
  TCP    127.0.0.1:49670        DESKTOP-0NRMEE7:49671  ESTABLISHED
  TCP    127.0.0.1:49671        DESKTOP-0NRMEE7:49670  ESTABLISHED
  TCP    127.0.0.1:49672        DESKTOP-0NRMEE7:49673  ESTABLISHED
  TCP    127.0.0.1:49673        DESKTOP-0NRMEE7:49672  ESTABLISHED
  TCP    192.168.0.135:1600     150.171.110.84:https   ESTABLISHED
  TCP    192.168.0.135:8306     24:https               ESTABLISHED
  TCP    192.168.0.135:12718    lcmaaa-as-in-f5:https  ESTABLISHED
  TCP    192.168.0.135:12722    52.123.253.161:https   TIME_WAIT
  TCP    192.168.0.135:13158    22:https               TIME_WAIT
  TCP    192.168.0.135:13641    whatsapp-cdn-shv-02-maa5:5222  ESTABLISHED
  TCP    192.168.0.135:14940    lcmaaa-as-in-f5:https  ESTABLISHED
  TCP    192.168.0.135:24331    104.18.39.21:https     ESTABLISHED
  TCP    192.168.0.135:24753    lcbomp-in-f188:5228    ESTABLISHED
  TCP    192.168.0.135:26510    150.171.110.84:https   ESTABLISHED
  TCP    192.168.0.135:26511    24:https               ESTABLISHED
  TCP    192.168.0.135:26512    172.66.2.5:http        ESTABLISHED
  TCP    192.168.0.135:26513    52.123.253.161:https   ESTABLISHED
  TCP    192.168.0.135:33265    202:4070               ESTABLISHED
  TCP    192.168.0.135:43393    35:https               ESTABLISHED
  TCP    192.168.0.135:49185    35:https               ESTABLISHED
  TCP    192.168.0.135:49433    4.213.25.241:https     ESTABLISHED
  TCP    192.168.0.135:56821    23.98.86.5:8883        ESTABLISHED
  TCP    192.168.0.135:56822    relay-8f76c373:https   ESTABLISHED
  TCP    192.168.0.135:65075    150.171.110.84:https   ESTABLISHED
```

```
C:\Users\Ramya>nslookup www.google.com
Server:  UnKnown
Address:  fe80::e21c:fcff:fee5:47c6

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482d:7700::
          2001:4860:4826:7700::
          2001:4860:482a:7700::
          2001:4860:4829:7700::
          2001:4860:4827:7700::
          2001:4860:4828:7700::
          2001:4860:482b:7700::
          2001:4860:482c:7700::
          142.251.156.119
          142.251.157.119
          142.251.150.119
          142.251.155.119
          142.251.152.119
          142.251.153.119
          142.251.151.119
          142.251.154.119
```

```
C:\Users\Ramya>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list        Loose source route along host-list.
    -h maximum_hops     Maximum number of hops to search for target.
    -i address          Use the specified source address.
    -n                  Do not resolve addresses to hostnames.
    -p period           Wait period milliseconds between pings.
    -q num_queries      Number of queries per hop.
    -w timeout          Wait timeout milliseconds for each reply.
    -4                  Force using IPv4.
    -6                  Force using IPv6.
```

```
C:\Users\Ramya>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3716:46e4:5f1a:704b%12
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fd01::3bdb:596d:9209:8711
   Temporary IPv6 Address. . . . . . : fd01::31fb:70ab:4a7f:5820
   Link-local IPv6 Address . . . . . : fe80::a4a1:9676:dd78:c212%16
   IPv4 Address. . . . . . . . . . . : 192.168.0.135
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1
```

```
C:\Users\Ramya>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

   -f             Clears the routing tables of all gateway entries.  If this is
                  used in conjunction with one of the commands, the tables are
                  cleared prior to running the command.

   -p             When used with the ADD command, makes a route persistent across
                  boots of the system. By default, routes are not preserved
                  when the system is restarted. Ignored for all other commands,
                  which always affect the appropriate persistent routes.

   -4             Force using IPv4.

   -6             Force using IPv6.

   command        One of these:
                    PRINT     Prints  a route
                    ADD       Adds    a route
                    DELETE    Deletes a route
                    CHANGE    Modifies an existing route
   destination    Specifies the host.
   MASK           Specifies that the next parameter is the 'netmask' value.
   netmask        Specifies a subnet mask value for this route entry.
                  If not specified, it defaults to 255.255.255.255.
   gateway        Specifies gateway.
   interface      the interface number for the specified route.
   METRIC         specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.
```

```
C:\Users\Ramya>ping google.com

Pinging google.com [142.250.205.110] with 32 bytes of data:
Reply from 142.250.205.110: bytes=32 time=4ms TTL=118
Reply from 142.250.205.110: bytes=32 time=3198ms TTL=118
Reply from 142.250.205.110: bytes=32 time=3ms TTL=118
Reply from 142.250.205.110: bytes=32 time=6ms TTL=118

Ping statistics for 142.250.205.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3198ms, Average = 802ms
```

10. ip

Syntax : ip address show

Output : 2 : enp0s0:

inet 192.108.1.100/24

11. mtr

Syntax : mtr google.com

Output : shows all hops to google.com with response time and packet loss

12. tcp dump

Syntax: tcpdump -i eth0

Output : listening on eth0

IP 192.168.1.105 8.8.8.8 : ICMP echo request

Result:

The various networking commands in linux and windows were executed successfully. Their Syntax purpose and outputs were studied and verified.